

Short Papers

Measuring Influence of Indices in DN Planning

Shubham Gupta , Vinod Kumar Yadav , *Senior Member, IEEE*, and Madhusudan Singh , *Senior Member, IEEE*

Abstract—The optimal distribution network (DN) setup in the modern framework baffles the DN planner, as various technoeconomic impacts need to be analyzed. The prevailing studies urge multiobjective problem formulation to quantify the wide range of technoeconomic performance indices; however, no adequate method is posited to measure the relative influence coefficient of these indices in DN planning. This article proposes a novel method based on the Shannon entropy formula to determine the relative influence level of indices in multiobjective optimally distributed generation unit(s) placement problems. Numerical results have been presented to validate the efficacy of the proposed approach over the previously published strategies.

Index Terms—Distribution network (DN), DN planning, Shannon entropy (SE), weighted-sum multiobjective analysis.

I. INTRODUCTION

The shifting from a vertically integrated structure to unbundling of power system brought fringe benefits but increased complexity in the planning, maintenance, and operational activities. In the new liberalized paradigm, the consumers have become prosumers as the traditional distribution networks (DNs) have now transcended to active DNs [1]. Renewable energy technologies (RETs), such as solar photovoltaic, wind, hydro, and others, are continuously improving and becoming cheaper and more efficient. Therefore, the integration of RETs in power DNs is escalating massively. However, due to the increased distributed generation (DG) penetration, the liabilities of the distribution network planner (DNP) are increasing as well [2]. The impact analysis of DGs in the power network has become essential so that the reliability and power availability do not get degraded. Unplanned and arbitrary DG placement may result in negative improvements in DNs' quantities, viz., a surge in power losses, system voltage drops, and diminished loading capability [3].

Conventionally, power loss reduction and voltage maximization are two common objectives for the DG placement problem. Finally, the extensive use of power electronics' devices in the DN has caused power quality and voltage fluctuation issues, raising concerns about finding new DG placement solutions for reliable and economical operations in a power DN [4]. The DG placement has been formulated as a multiobjective problem (MOP) to take different technical and economic viewpoints simultaneously. The stud krill herd optimization technique is used in [5] to solve a weighted-sum MOP for the ideal location and sizing of DGs, where the appropriate weight preferences for the problem objectives have been taken subjectively. An analytical hierarchical process (AHP) technique is widely utilized for obtaining the weight

coefficients in the weighted-sum method of MOP [6], [7], [8], [9]. However, the pairwise comparison matrix in the AHP technique is formed based on the decision maker's choice. The weighted normalized decision matrix is constructed based on predetermined weight coefficients in the technique for order preference by similarity to ideal solution to find the preferred alternative most near to the positive ideal solution while solving MOP for optimal siting and sizing of DGs in [9] and [10]. The selection of appropriate weight factors in weighted-sum MOP is still challenging and poses ambiguity [11].

Various methods have recently been developed based on the Pareto dominance concept to obtain tradeoff solutions among the different objectives in a MOP, such as Harris Hawks optimizer [11], nondominated sorting genetic algorithm [12], and multiobjective differential evolution [13]. However, relatively fewer efforts have been laid into presenting strategies for selecting the best tradeoff from a set of nondominated solutions, which still requires further research. Often, distance-based methods, such as gray relational analysis (GRA) [11] and fuzzy decision [12], [13], [14], are utilized instead of investigating the inherent relationship (i.e., relative influence) among objectives to find the best alternative.

Finally, it can be concluded that the relative influence coefficients (RICs) of the investigated indices in MOP formulations for optimal placement of DG are usually assessed empirically in previous studies, although it is worth noting that the impact of an objective (or indices) varies with the change in the network's physical properties. Therefore, achieving pragmatic solutions seems deceptive without impartially evaluating the RICs of indices in a MOP. This article presents a novel method based on the Shannon entropy (SE) formula to evaluate the relative influence of each considered index objectively and effectively in DN planning.

II. PROBLEM FORMULATION

In generic, a weighted-sum multiobjective optimization problem for DG placement in DN can be written as follows:

$$\text{minimize } \sum_{i=1}^N \xi_i * f_i(x) \quad (1)$$

$$\text{s.t. } h(x) = 0 \quad (2)$$

$$g(x) \geq 0 \quad (3)$$

where N in (1) is the total number of objectives considered. In this article, ξ_i is referred to as the RIC representing the degree of importance of the i th objective function relatively in MOP formulation. In (2) and (3), the equality and inequality constraints represent the power flow balance, active and reactive power limits, voltage limits, and other operational and security bounds, as described in (4)–(8). In (1), ξ_i of each considered objective is usually based on the decision maker's choice, resulting in bias. This article proposed a new strategy based

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The authors are with the Department of Electrical Engineering, Delhi Technological University, Delhi 110042, India (e-mail: shubham.gupta6691@gmail.com; vinodkumar@dtu.ac.in; madhusudan@dce.ac.in).

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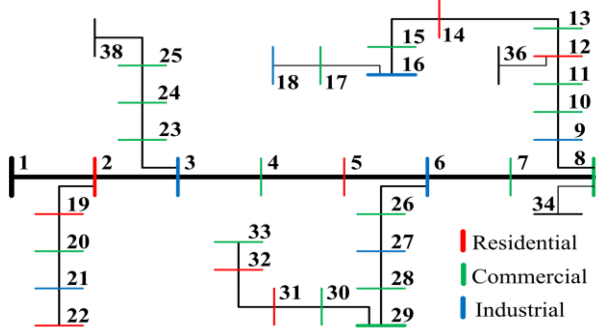


Fig. 1. Single line diagram of 38-node test system depicting residential, commercial, and industrial loads.

on the SE formula for assessing the RIC of each objective function equitably in a multiobjective optimally DG placement problem.

The DN power flows' equations [15] and DG operating constraints to be satisfied $\forall f \in \Omega_n \forall l = ft \in \Omega_L$, and $\forall i \in \Omega_{DG}$ are

$$\begin{aligned} -P_{inj,t} &= -(P_{g,t} - P_{d,t}) \\ &= P_{ft} - \frac{r_{ft}}{V_f^2} \{P_{ft}^2 + Q_{ft}^2\} - \sum_{c:t \rightarrow c} P_{tc} \end{aligned} \quad (4)$$

$$-Q_{inj,t} = Q_{d,t} = Q_{ft} - \frac{x_{ft}}{V_f^2} \{P_{ft}^2 + Q_{ft}^2\} - \sum_{c:t \rightarrow c} Q_{tc} \quad (5)$$

$$\begin{aligned} V_t^2 &= V_f^2 + \frac{r_{ft}^2 + x_{ft}^2}{V_f^2} \{P_{ft}^2 + Q_{ft}^2\} \\ &\quad - 2 \{r_{ft} P_{ft} + x_{ft} Q_{ft}\} \end{aligned} \quad (6)$$

$$V_f^{\min} \leq V_f \leq V_f^{\max} \quad (7)$$

$$P_{g,i}^{\min} \leq P_{g,i} \leq P_{g,i}^{\max} \quad (8)$$

where $P_{inj,t}$ and $Q_{inj,t}$ in (4) and (5) are the net injected real and reactive power to the t th bus, respectively. $P_{d,t}$ and $Q_{d,t}$ are the active and reactive power demands at the t th bus, respectively. P_{ft} and Q_{ft} are the active and reactive power flow between buses f and t , respectively. V_f , $V_f^{\min}$, and $V_f^{\max}$ in (7) are the voltage magnitude and lower/upper limits of V of the f th bus, respectively. r_{ft} and x_{ft} are the resistance and reactance of the line connecting buses f and t , respectively. $P_{g,i}$ in (8) is the active power generated by the connected DG at the i th bus.

III. SE INCORPORATING DNP PRIORITY INFORMATION (PI)

The SE is the most practiced method in information theory to measure the value dispersion in multicriteria decision making [16]. The fundamental principle of the SE is that the greater the dispersal in the measured value, the greater the differentiation of the criteria; as a result, more information can be obtained, moreover, the higher the importance of that criteria, and vice-versa. The SE formula is expressed as follows:

$$e_i = -e_0 \sum_{k=1}^s \bar{I}_{ki} \cdot \ln \bar{I}_{ki} \quad (9)$$

here e_0 is termed as the entropy constant and is equal to $(\ln s)^{-1}$, $\bar{I}_{ki}$ is the normalized k th value of the i th index. From (9), the RI coefficient can be measured as follows:

$$\xi_i = \frac{(1 - e_i)}{\sum_{i=1}^N (1 - e_i)}. \quad (10)$$

TABLE I
DN INDICES CONSIDER FOR DG PLACEMENT PROBLEM

DN Indices		Mathematical Notation
Technical Indices [17]	APLI	$f_1 = 1 - \frac{\left(\sum_{t \in U_t} P_{Loss}^{k,t} \right)}{\left(\sum_{t \in U_t} P_{Loss}^{0,t} \right)}$
	Reactive Power Loss Index	$f_2 = 1 - \frac{\left(\sum_{t \in U_t} Q_{Loss}^{k,t} \right)}{\left(\sum_{t \in U_t} Q_{Loss}^{0,t} \right)}$
	MVDI	$f_3 = 1 - \frac{\sum_{b \in U_b} (\bar{V}_o - \bar{V}_b^k / \bar{V}_o)^4}{\sum_{b \in U_b} (\bar{V}_o - \bar{V}_b^0 / \bar{V}_o)^2}$
	LCLI	$f_4 = 1 - \frac{1}{T} \sum_{t \in U_t} \max \left(\left \bar{S}_t^{k,t} \right / \left \bar{S}_t^{\max} \right \right)^{t \in U_t}$
Economic Index	Annual Equivalent Cost Index	$f_5 = 1 - \left(\sum_{t \in U_t} C_{cost}^{k,t} / \sum_{t \in U_t} C_{cost}^{0,t} \right)$ where $C_{cost}^k = \left(P_{DG}^k \times C_I^{DG} \times CRF \right) + \left(P_{DG}^k \times C_{OM}^{DG} \right) + \left(C_E \left(P_{Loss}^k + P_L \right) \right)$
	Average Energy Not Supplied Index	$f_6 = 1 - \left(\sum_{q=1}^L P_q^k \times U_q / \sum_{q=1}^L P_q^0 \times U_q \right)$
Reliability Index [19]		

In some given conditions, the DNP strategically sets a preference for an index; thus, RI coefficients must be changed accordingly. This PI can be incorporated by modifying the ξ_i in (10) for “ n ” indices with PI as in (11) and “ m ” indices without PI as in (12).

$$\xi_{n \in N} = \frac{(1 - e_n) + \left(\zeta_n * \sum_{i=1}^N (1 - e_i) \right)}{\sum_{i=1}^N (1 - e_i)} \quad (11)$$

$$\xi_{m \in \{N \setminus n\}} = \frac{(1 - e_m)}{\sum_{i=1}^N (1 - e_i)} - \frac{\sum_{n \in N} \zeta_n * (1 - e_m)}{\sum_{i=1}^N (1 - e_i) - \sum_{n \in N} (1 - e_n)} \quad (12)$$

The AHP technique provides flexibility to customize the weights associated with indices in a MOP, which is mainly useful for subjective analysis [17]. On the contrary, DN planning is a type of complex problem where the RICs of indices need to be assessed objectively based on their physical significance, which can be handled very efficiently by the proposed method. Furthermore, it allows the incorporation of PI given by the DNP in certain situations.

IV. SOLUTION METHODOLOGY

Finally, numerous methodologies have been suggested for the optimal placement of DG in power DNs, and in them, many methods focus on multiobjective optimization. However, no adequate strategy is directed to quantify the RIC of each objective resulting in a chance of getting a pseudo-optimal solution. Therefore, this article presents a novel approach based on the SE formula to solve the multiobjective optimal DG placement problem. The SE formula relatively quantifies the degree of importance of indices in formulated MOP. The proposed approach is solved in the following steps.

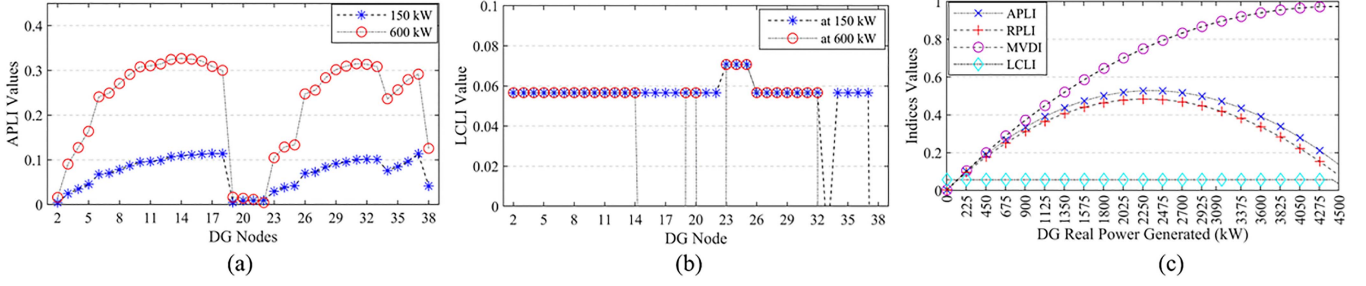


Fig. 2. Technical indices scores for a 38-node test system in different DG placement problems. (a) APLI scores. (b) LCLI scores. (c) Indices scores at sixth bus.

TABLE II
SIMULATION RESULTS FOR ALL FOUR CONSIDERED CASES

Cases	RI Coefficients						PI	(ζ_i)	Optimal Solution	DN Quantities				
	ξ_1	ξ_2	ξ_3	ξ_4	ξ_5	ξ_6				APL (kW)	RPL (kVar)	V_{min} (pu.)	AEC (MS)	AENS
Base	-	-	-	-	-	-	-	-	-	152.80	101.50	18/0.9252	-	1.6732
Case-1	0.24	0.27	0.10	0.39	-	-	-	-	Sixth bus/2523 kW	72.23	52.74	18/0.9616	1.718	1.6236
Case-2	0.34	0.23	0.09	0.34	-	-	$\zeta_1 = 10\%$	-	Sixth bus/2480 kW	72.06	52.58	18/0.9611	1.724	1.6227
Case-3	0.29	-	-	-	0.34	0.37	$\zeta_1 = 10\%$	-	Sixth bus/2834 kW	75.04	54.92	18/0.9660	1.678	1.6165

TABLE III
COMPARISON AMONG DIFFERENT EXISTING METHODS AND PROPOSED APPROACH

Methods	AHP	GRA*	FDM*	Proposed Approach
RIC/ Preference Weights	ξ_1 ξ_5 ξ_6	0.54 0.16 0.30	- - -	0.19 0.39 0.42
Optimal Solution	Sixth bus/ 2492	Sixth bus/ 1995	Sixth bus/ 1995	Sixth bus/ 3180
APL (kW)	72.11	73.52	73.52	81.09
RPL (kW)	52.63	52.95	52.95	59.22
V_{min} (p.u.)	18/0.9613	18/0.9543	18/0.9543	18/0.9707
AEC (MS)	1.722	1.788	1.788	1.636
AENS	1.6234	1.6332	1.6332	1.6096

* GRA and FDM strategies determine the best-compromised solution from a set of nondominated solutions of Pareto-front.

- 1) Gather the input data and solve the load flow problem for the base case without the placement of DG for comparison and analysis of results with the proposed approach.
- 2) Define the objectives of MOP and evaluate their RIC using the SE-based strategy, as described in Section III, satisfying the constraint as given in the following:

$$\sum_{i=1}^N \xi_i = 1. \quad (13)$$

- 3) Formulate the weighted-sum multiobjective optimal DG placement problem and obtain the optimal solution

$$\text{maximize} \quad \sum_{i=1}^N \xi_i * f'_i(x) \quad (14)$$

$$\text{s.t.} \quad (4) - (8). \quad (15)$$

In (14), the problem objectives $f'_i(x)$ are reconstructed to model them unitless for the relative measurement of RIC, where the

$f'_i(x)$ is elucidated as the index in this article

$$f'_i = 1 - \left(\frac{f_i^k(x)}{f_{base}^k(x)} \right). \quad (16)$$

The formulated problem is a mixed-integer programming problem that simultaneously finds the optimal location and size of the DG in DN planning and could be solved through any classical or nature-inspired multiobjective optimization technique. This work primarily focuses on objectively determining the relative influence of indices in DN planning.

V. CASE STUDY

The proposed method is examined on a 38-node test system [18], consisting of the total active and reactive power demand of 3501.6 kW and 1870.2 kVar, respectively. The system configuration has different types of loads, viz., industrial, commercial, and residential, as shown in Fig. 1. For the simulation, the base quantities S_{base} and kV_{base} are taken for analysis as 100 MVA and 12.66 kV, respectively. Considering many indices for measuring the alteration in the network quantities during the DG placement problem analysis is a colossal need to tend toward the practical application. This work considers four technical indices, one economic and one reliability index, to measure their influence on DN planning while formulating the DG placement problem. Mathematically, these indices are expressed in Table I.

A. Influence of DN Indices on DG Placement Problem

It is a well-known fact that the DG placement alters the DN quantities, viz., change in real and reactive power losses, reshaping the voltage profile, and regulating the line capacity, which can be elucidated through Fig. 2(a)–(c). The greater the degree of dispersion in DN quantity, the higher the impact of the designated index in the MOP problem. From Fig. 2(a) and (b), it can be observed that the active power loss index (APLI) scores are more dispersed compared with line capacity limit index (LCLI) scores when one DG of 150 kW is placed in DN. Hence, APLI relatively influences more the MOP problem than LCLI; therefore, RIC should be more for APLI. On the contrary, when a DG of size 600 kW is placed, the line capacity limit is violated

for many configurations; therefore, priority should be given to LCLI, and the RIC of LCLI should be put higher than to APLI. Fig. 2(c) represents the various indices value when DG size is varied for a node location (i.e., sixth bus). In Fig. 2(c), the LCLI scores barely changed; in contrast, the maximum voltage deviation index (MVDI) shows the highest dispersion among all indices.

B. Simulation Results of the Proposed Approach

The following four cases are simulated and analyzed for the DG placement problem to illustrate the effectiveness of the proposed approach.

- Case-1: Optimal placement and sizing of DG, considering technical indices (f_1, f_2, f_3 , and f_4).
- Case-2: Optimal placement and sizing of DG, considering technical indices (f_1, f_2, f_3 , and f_4) and index PI.
- Case-3: Optimal placement and sizing of DG, considering technical, economic, and reliability indices (f_1, f_5 , and f_6) and index PI.

The problem formulation is simulated for the placement of a single DG in DN; however, the proposed approach can obtain the solution for multiple DGs in both simultaneous or sequential placement manner.

C. Comparison Study

This article compares the proposed methodology with the previously published methods, i.e., AHP, GRA, and fuzzy decision method (FDM), to validate the efficacy. The comparison results are reported in Table III. The optimal DG size and location are assessed considering the active power loss (f_1), cost factor (f_5), and reliability factor (f_6) indices.

The prevailing studies did not articulate the fact that the RICs differ as the problem formulation changes for the DG placement problem, which can be deduced from the simulated results in Tables II and III. On this basis, it could be said that the proposed approach can produce an objective and comprehensive solution with physical significance.

VI. CONCLUSION

A novel framework to measure the influence of indices on the DG placement problem during DN planning is proposed, which utilizes the SE formula to quantify the relative degree of importance objectively and effectively. The main advantages of the proposed method are that incorporates the index PI given by DNP in inevitable situations and is easily executable, and fit in with other similar multiobjective optimization problem.

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